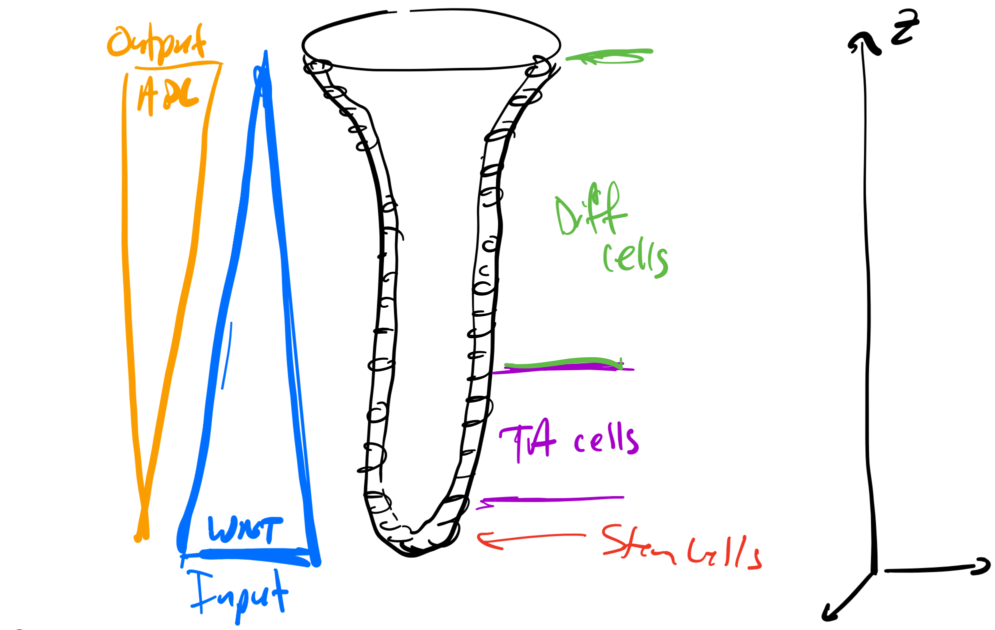
Single Cell

$$W \longrightarrow \begin{cases} APC \\ \beta\text{-cat} \\ Axin \end{cases} \qquad \text{ODE system}$$

Crypt level:

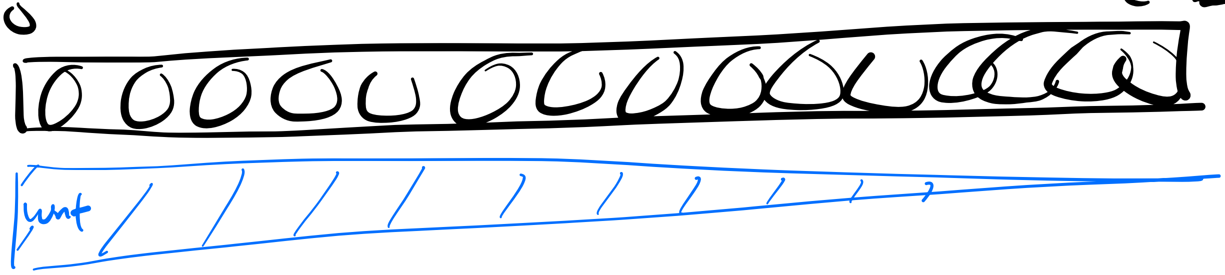


Bottom

$z=0$

Top

$z=L$

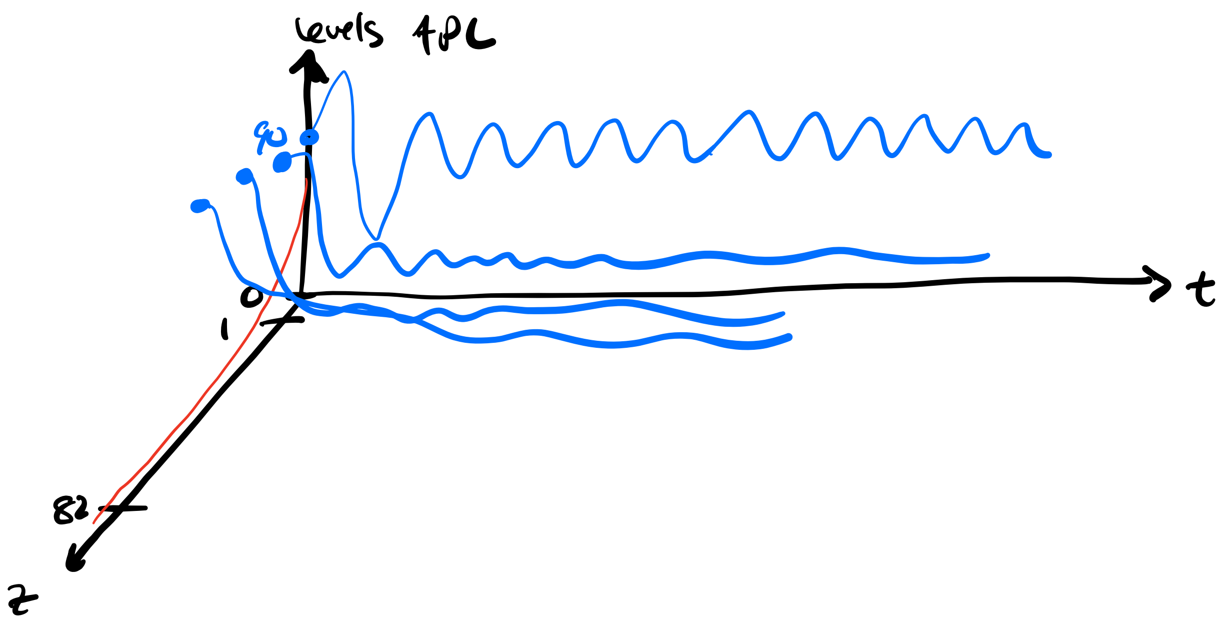


$$q(t,z) = \text{density of cells (cells per unit length)} = \frac{\text{cells}}{\text{length}}$$

$$n(t) = \int_0^L q(t,z)\,dz = \text{cells}$$

$\underline{z=0}$: $\underline{w=1}$: $\begin{cases} A(t,0) \\ B(t,0) \\ X(t,0) \end{cases}$

$\underline{z=0.1}$: $w=.98$ $\begin{cases} A(t,0.1) \\ B(t,0.1) \\ X(t,0.1) \end{cases}$



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